CHANDIGARH UNIVERSITY

Gharuan, Mohali, Punjab

MINI PROJECT REPORT

Netflix Movies and TV Shows Analysis

Exploratory Data Analysis of Genres, Ratings and Release Years using Pandas, NumPy, Matplotlib and Seaborn

Subject Name	Python Programming
Subject Code	25CAH-221
Submitted By	Hardeep Singh Sodhi
UID	25bcd10215
Class	25BCD-2(A)
Submitted To	Ms. Preeti Verma

Table of Contents

1. Introduction

2. Objectives

3. Tools and Libraries Used

4. Dataset Description

5. Data Cleaning and Preparation

6. Exploratory Data Analysis

   6.1 Movies vs TV Shows

   6.2 Genre Analysis

   6.3 Rating Analysis

   6.4 Release Year Analysis

   6.5 Duration and Year Added (extra)

7. Key Findings

8. Conclusion

9. Future Scope

10. References

Appendix: Complete Python Code

1. Introduction

Netflix is one of the largest streaming platforms in the world, offering thousands of movies and TV shows from many countries. Every title on the platform carries information such as its type, genres, audience rating, release year, country and duration. When this information is studied with the help of programming, it can show what kind of content a platform focuses on and how that focus has changed over time.

This mini project performs an **Exploratory Data Analysis (EDA)** on the Netflix Movies and TV Shows dataset using Python. The analysis concentrates on three main aspects: **genres**, **ratings** and **release years**. Data handling is done with Pandas and NumPy, and the results are presented through Matplotlib and Seaborn visualisations.

2. Objectives

- To load and understand the Netflix titles dataset using Pandas.
- To clean the data by handling missing values and converting columns to suitable formats.
- To find which genres are the most common, separately for movies and TV shows.
- To study the distribution of content ratings and the audience groups they represent.
- To analyse how the number of titles changes with release year and release decade.
- To use NumPy for basic statistics such as mean, median, standard deviation and percentiles.
- To present the findings through clear charts made with Matplotlib and Seaborn.

3. Tools and Libraries Used

Tool / Library	Purpose in this project
Python 3	Programming language used for the whole analysis
Pandas	Reading the CSV file, cleaning, grouping, exploding genres, cross-tabulation
NumPy	Numerical statistics (mean, median, standard deviation, percentiles) on duration and years
Matplotlib	Base plotting library for figures, titles, labels and saving images
Seaborn	Statistical charts such as count plots, bar plots, histograms and heatmaps
Jupyter / VS Code	Environment for writing and running the code

4. Dataset Description

The dataset used is the **Netflix Movies and TV Shows** dataset by Shivam Bansal (Kaggle), collected from the Flixable search engine and listing titles available on Netflix up to early 2021. The file *netflix_titles.csv* contains **7,787 rows** and **12 columns**. Each row is one title.

Column	Description
show_id	Unique ID of the title
type	Movie or TV Show
title	Name of the title
director	Director(s) of the title
cast	Main actors
country	Country or countries of production
date_added	Date on which it was added to Netflix
release_year	Year in which the title was originally released
rating	Audience maturity rating (TV-MA, PG-13, etc.)
duration	Minutes for movies, number of seasons for TV shows
listed_in	Genres / categories (comma separated)
description	Short summary of the title

Loading the data

```
import pandas as pd, numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("netflix_titles.csv")
print(df.shape)        # (7787, 12)
df.info()
df.head()
```

5. Data Cleaning and Preparation

Real-world data is rarely perfect, so the first step was to check missing values and duplicates. The check showed that no duplicate rows were present (0 duplicates). Missing values were found in five columns, as shown below.

Column	Missing values	Action taken
director	2389	Left as it is (not used in this analysis)
cast	718	Left as it is (not used in this analysis)
country	507	Filled with 'Unknown'
date_added	10	Converted to datetime; invalid values become NaT
rating	7	Filled with 'Unknown'

```
print(df.isnull().sum())
print(df.duplicated().sum())          # 0

df["rating"]  = df["rating"].fillna("Unknown")
df["country"] = df["country"].fillna("Unknown")
df["date_added"] = pd.to_datetime(df["date_added"].str.strip(), errors="coerce")
df["year_added"] = df["date_added"].dt.year
df["decade"]     = df["release_year"] // 10 * 10

# a title can belong to several genres -> split and explode
df["genres"] = df["listed_in"].str.split(", ")
g = df.explode("genres").reset_index(drop=True)
```

The *listed_in* column stores several genres in one cell, for example *'International TV Shows, TV Dramas, TV Sci-Fi & Fantasy'*. To count each genre properly, the text was split on the comma and then expanded with **explode()** so that every genre gets its own row. Because of this, the genre counts in the next section add up to more than the number of titles, as one title can have more than one genre.

6. Exploratory Data Analysis

6.1 Movies vs TV Shows

Out of 7,787 titles, **5,377 (69.1%) are movies** and **2,410 (30.9%) are TV shows**. Movies are therefore more than twice as common as TV shows in this dataset.

```
sns.countplot(data=df, x="type", palette=["#E50914", "#221f1f"])
plt.title("Movies vs TV Shows on Netflix")
plt.show()
```

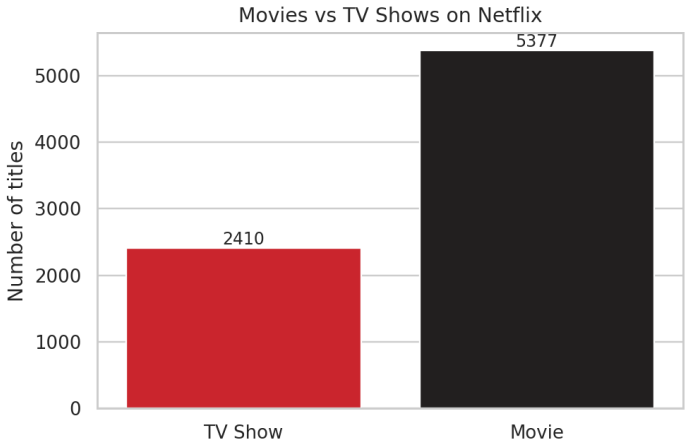


Figure 1: Count of Movies and TV Shows

6.2 Genre Analysis

After splitting the *listed_in* column, the dataset was found to contain **42 unique genres**. The most frequent genre is **International Movies** (2,437 titles), followed by **Dramas** (2,106) and **Comedies** (1,471). The top ten genres are shown in Figure 2.

```
genre_counts = g["genres"].value_counts()
print(genre_counts.head(10))

sns.barplot(x=genre_counts.head(10).values,
            y=genre_counts.head(10).index, palette="Reds_r")
plt.title("Top 10 Genres")
```

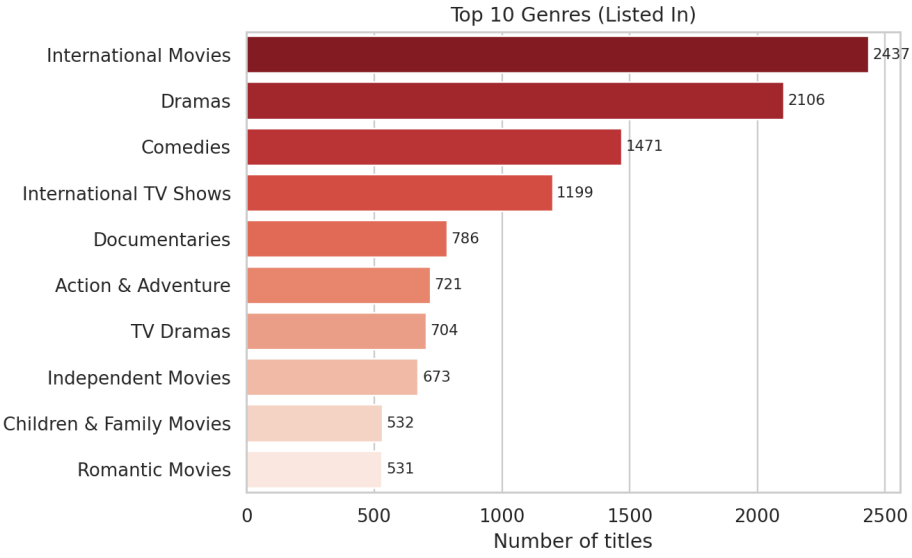


Figure 2: Top 10 genres on Netflix

Since movies and TV shows use different genre labels, the two types were also compared separately. Among movies, **International Movies**, **Dramas** and **Comedies** lead the list. Among TV shows, **International TV Shows**, **TV Dramas** and **TV Comedies** are the top three, with Crime TV Shows and Kids' TV also strong.

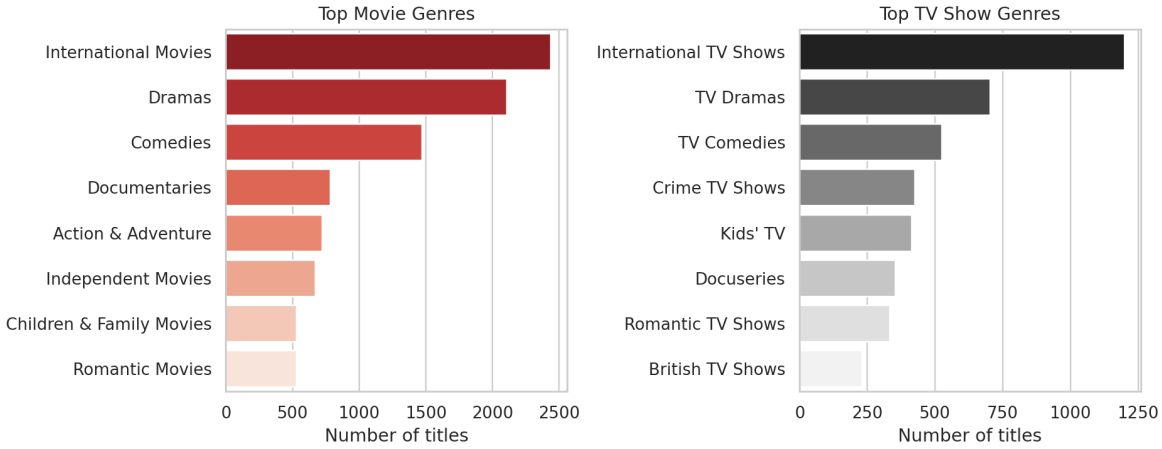


Figure 3: Top genres for movies (left) and TV shows (right)

To see how genre popularity has moved over time, a heatmap of the top eight genres against release decade was drawn (Figure 4). Almost every genre has its highest count in the 2010s, which matches the large growth in Netflix's catalogue during that period.

```
top8 = genre_counts.head(8).index
g2 = g[g["genres"].isin(top8) & (g["release_year"] >= 1990)].copy()
g2["decade"] = g2["decade"].astype(int).astype(str) + "s"
heat = pd.crosstab(g2["genres"], g2["decade"]).loc[top8]
sns.heatmap(heat, annot=True, fmt="d", cmap="Reds", linewidths=.5)
```

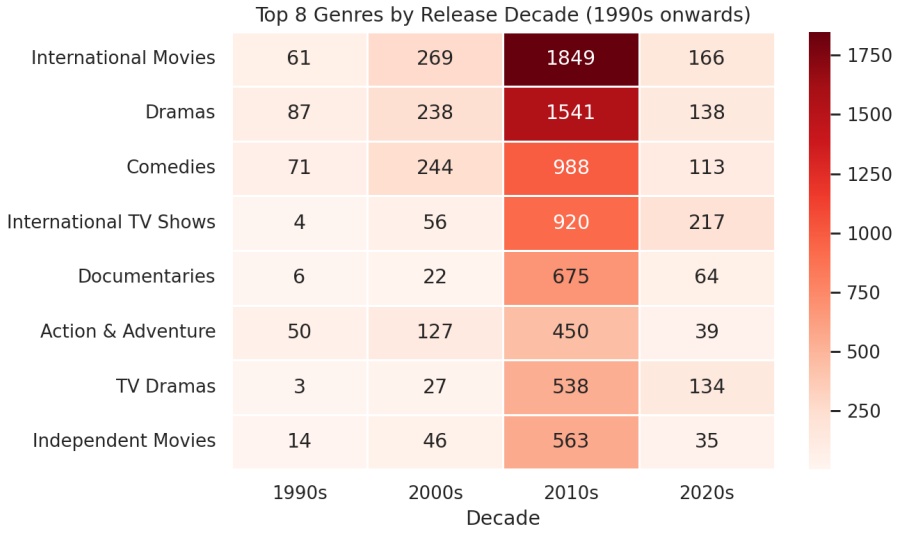


Figure 4: Genre vs release decade heatmap

6.3 Rating Analysis

The rating column shows the maturity level of a title. **TV-MA** is the most common rating with 2,863 titles (36.8%), followed by **TV-14** with 1,931 titles (24.8%). Together these two adult/teen ratings cover about 61.6% of all content. Child-friendly ratings such as TV-Y, TV-Y7 and TV-G are much smaller in number.

```
rating_counts = df["rating"].value_counts()
sns.barplot(x=rating_counts.index, y=rating_counts.values, palette="Reds_r")
plt.xticks(rotation=45)
```

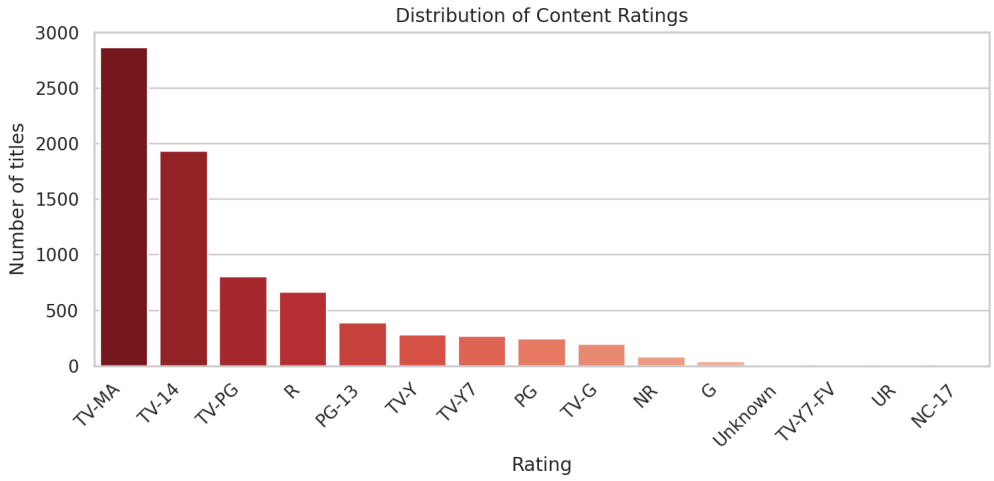


Figure 5: Distribution of content ratings

To make the ratings easier to interpret, they were grouped into audience categories: **Adults** (TV-MA, R, NC-17), **Teens** (TV-14, PG-13), **Older Kids** (TV-PG, PG) and **Kids** (TV-Y, TV-Y7, TV-G, G).

Audience group	Number of titles	Share (%)
Adults	3,531	45.3
Teens	2,317	29.8
Older Kids	1,053	13.5
Kids	790	10.1
Other/Unrated	96	1.2

```
mp = {"TV-MA":"Adults","R":"Adults","NC-17":"Adults","TV-14":"Teens","PG-13":"Teens",
      "TV-PG":"Older Kids","PG":"Older Kids","TV-Y7":"Kids","TV-Y7-FV":"Kids","TV-Y":"Kids","TV-G":"Kids","G":"Kids
df["audience"] = df["rating"].map(mp).fillna("Other/Unrated")
df["audience"].value_counts()
```

The rating distribution was also split by content type. For TV shows, TV-MA and TV-14 dominate, while movies show a wider spread that includes R and PG-13.

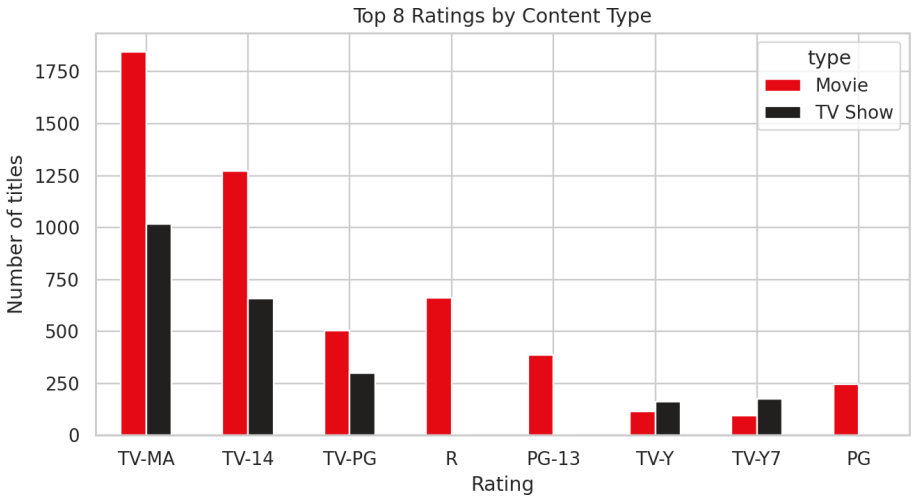


Figure 6: Top ratings by content type

6.4 Release Year Analysis

Titles in the dataset were released between **1925** and **2021**. NumPy was used to compute basic statistics: the mean release year is **2013.9** and the median is **2017**. Since the median is higher than the mean, a few very old titles pull the average down, and most content is recent. The year with the most titles is **2018** with 1,121 titles, and about **70.0%** of all titles were released in 2015 or later.

```
ry = df["release_year"].values
print(np.mean(ry), np.median(ry), ry.min(), ry.max())
year_counts = df["release_year"].value_counts().sort_index()
print(year_counts.idxmax(), year_counts.max())       # 2018 1121
print((df["release_year"] >= 2015).mean() * 100)     # 70.0
```

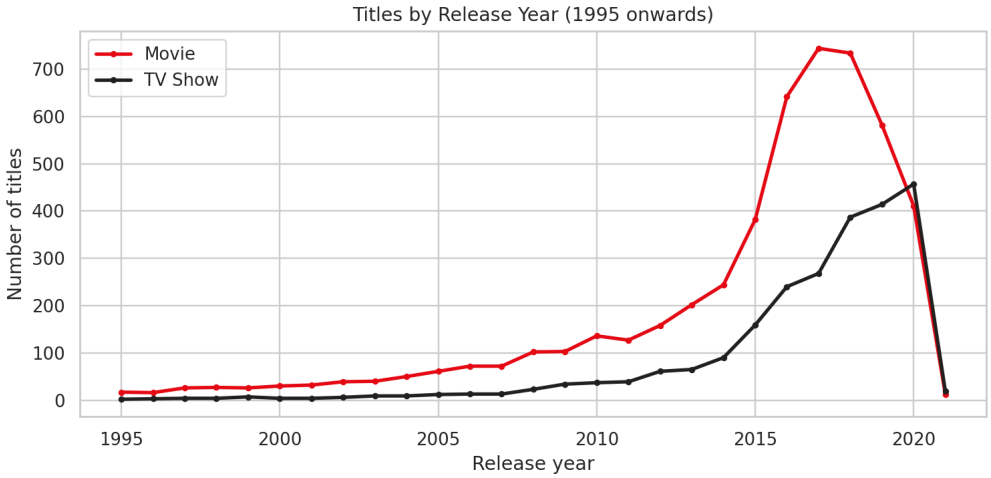


Figure 7: Number of titles by release year, split by type

The line chart shows a sharp rise in the number of movies and TV shows released after 2010. Movie releases grow faster and peak around 2017-2018, while TV shows grow more steadily. The fall after 2018 is partly because the dataset was collected in early 2021, so the most recent years are not complete.

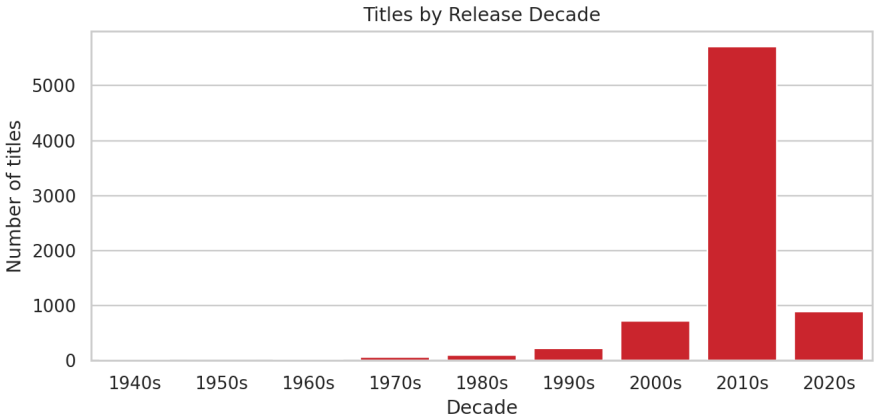


Figure 8: Titles by release decade

Decade	Titles
1940s	14
1950s	11
1960s	25
1970s	67
1980s	106
1990s	225
2000s	728
2010s	5,711
2020s	899

6.5 Duration and Year Added (additional analysis)

For movies, the duration column holds minutes. After extracting the number, NumPy gave a mean of **99.3 minutes**, a median of **98 minutes** and a standard deviation of **28.5 minutes**. Half of all movies lie between 86 and 114 minutes (25th to 75th percentile). The shortest movie is 3 minutes and the longest is 312 minutes.

```
mv = df[df["type"] == "Movie"].copy()
mv["mins"] = mv["duration"].str.extract(r"(\d+)").astype(float)
d = mv["mins"].dropna().values
print(np.mean(d), np.median(d), np.std(d))
print(np.percentile(d, [25, 75]))
sns.histplot(d, bins=30, color="#E50914")
```

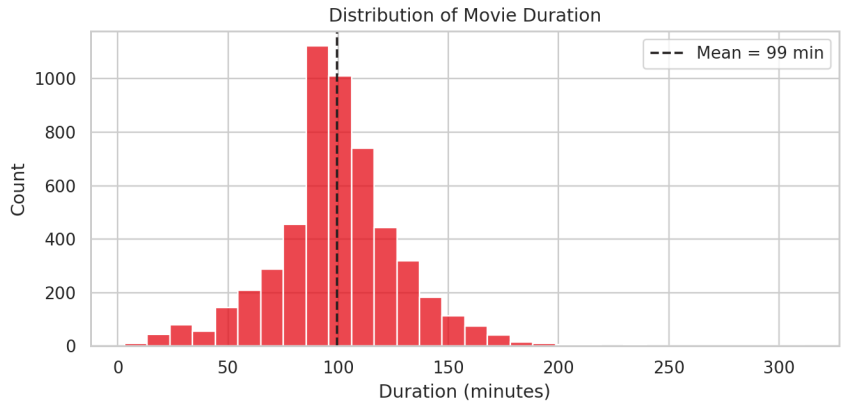


Figure 9: Distribution of movie duration

For TV shows, **66.7%** have only one season, which shows that most series on Netflix are limited or recently launched. The year in which titles were added to Netflix also shows strong growth: additions rose from only 88 in 2015 to **2,153 in 2019**. The figure for 2021 is small only because the data stops in January 2021.

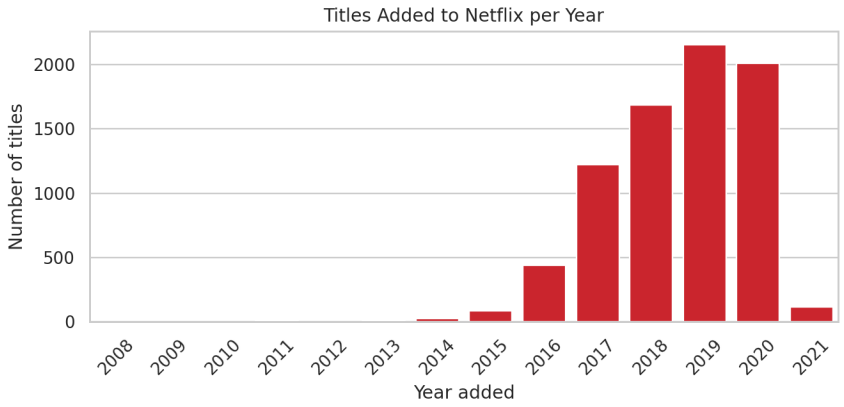


Figure 10: Titles added to Netflix per year

7. Key Findings

- Movies form about 69% of the catalogue and TV shows about 31%.
- **Genres:** International Movies (2,437), Dramas (2,106) and Comedies (1,471) are the three most common genres. International content is strong for both movies and TV shows.
- **Ratings:** TV-MA and TV-14 together make up about 62% of titles, so Netflix content is aimed mostly at adult and teenage viewers. Adult-rated content alone is 45%.
- **Release years:** About 70% of titles were released in 2015 or later, and 73% belong to the 2010s. The peak release year is 2018.
- **Duration:** The typical movie is close to 98 minutes long, and 66.7% of TV shows have a single season.
- **Growth:** The number of titles added each year grew very fast between 2015 and 2019.
- **Countries:** The United States (3,296) and India (990) contribute the most titles, followed by the United Kingdom, Canada and France.

8. Conclusion

This project showed how Python libraries can be combined to analyse a real-world dataset from start to finish. Pandas made it easy to load, clean and reshape the Netflix data, NumPy provided quick statistics, and Matplotlib with Seaborn turned the numbers into readable charts.

The analysis shows that Netflix's catalogue is dominated by movies, with International Movies, Dramas and Comedies as the leading genres. Most titles carry mature ratings (TV-MA and TV-14), and the large majority were released after 2010, which reflects the rapid expansion of the platform in recent years. The project also helped in understanding important EDA ideas such as handling missing values, splitting multi-valued columns with *explode()*, grouping, cross-tabulation and heatmaps.

9. Future Scope

- Combine this dataset with IMDb ratings to study how genres and ratings relate to audience scores.
- Analyse the *cast* and *director* columns to find the most frequent actors and directors.
- Use text analysis (word clouds, TF-IDF) on the *description* column.
- Build a simple content-based recommendation system using genres and descriptions.
- Create an interactive dashboard with Plotly or Streamlit.

10. References

- Shivam Bansal, *Netflix Movies and TV Shows* dataset, Kaggle. https://www.kaggle.com/datasets/shivamb/netflix-shows
- TidyTuesday (R for Data Science community), Netflix Shows, 2021-04-20 (copy of the dataset used here).

- Pandas documentation. https://pandas.pydata.org/docs/
- NumPy documentation. https://numpy.org/doc/
- Matplotlib documentation. https://matplotlib.org/stable/
- Seaborn documentation. https://seaborn.pydata.org/
- Wes McKinney, *Python for Data Analysis*, O'Reilly Media.

Appendix: Complete Python Code

```
import pandas as pd, numpy as np, matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt, seaborn as sns, json
sns.set_theme(style="whitegrid", font_scale=1.0)
RED="#E50914"; DARK="#221f1f"
df = pd.read_csv("netflix_titles.csv")
S = {}
S["rows"], S["cols"] = df.shape
S["missing"] = df.isna().sum().to_dict()
S["dups"] = int(df.duplicated().sum())

# cleaning
df["rating"] = df["rating"].fillna("Unknown")
df["country"] = df["country"].fillna("Unknown")
df["date_added"] = pd.to_datetime(df["date_added"].str.strip(), errors="coerce")
df["year_added"] = df["date_added"].dt.year
df["decade"] = (df["release_year"]//10*10)
df["genres"] = df["listed_in"].str.split(", ")
S["type_counts"] = df["type"].value_counts().to_dict()

# 1 type
fig, ax = plt.subplots(figsize=(6,4))
sns.countplot(data=df, x="type", palette=[RED, DARK], ax=ax)
for p in ax.patches:
    ax.annotate(int(p.get_height()), (p.get_x()+p.get_width()/2, p.get_height()), ha="center",
    va="bottom", fontsize=10)
ax.set_title("Movies vs TV Shows on Netflix"); ax.set_xlabel(""); ax.set_ylabel("Number of titles")
plt.tight_layout(); plt.savefig("f1_type.png", dpi=170); plt.close()

# genres
g = df.explode("genres").reset_index(drop=True)
gc = g["genres"].value_counts()
S["n_genres"] = int(gc.shape[0])
S["top_genres"] = gc.head(10).to_dict()
fig, ax = plt.subplots(figsize=(8,5))
sns.barplot(x=gc.head(10).values, y=gc.head(10).index, palette="Reds_r", ax=ax)
for i,v in enumerate(gc.head(10).values): ax.text(v+15, i, str(v), va="center", fontsize=9)
ax.set_title("Top 10 Genres (Listed In)"); ax.set_xlabel("Number of titles"); ax.set_ylabel("")
plt.tight_layout(); plt.savefig("f2_genres.png", dpi=170); plt.close()

# genres per type
gm = g[g.type=="Movie"]["genres"].value_counts().head(8)
gt = g[g.type=="TV Show"]["genres"].value_counts().head(8)
S["top_movie_genres"] = gm.to_dict(); S["top_tv_genres"] = gt.to_dict()
fig, axs = plt.subplots(1,2, figsize=(11,4.5))
sns.barplot(x=gm.values, y=gm.index, palette="Reds_r", ax=axs[0]); axs[0].set_title("Top Movie Genres")
sns.barplot(x=gt.values, y=gt.index, palette="Greys_r", ax=axs[1]); axs[1].set_title("Top TV Show
    Genres")
for a in axs: a.set_xlabel("Number of titles"); a.set_ylabel("")
plt.tight_layout(); plt.savefig("f3_genres_type.png", dpi=170); plt.close()

# ratings
rc = df["rating"].value_counts()
S["rating_counts"] = rc.to_dict()
order = rc.index.tolist()
fig, ax = plt.subplots(figsize=(9,4.5))
sns.barplot(x=rc.index, y=rc.values, order=order, palette="Reds_r", ax=ax)
ax.set_title("Distribution of Content Ratings"); ax.set_xlabel("Rating"); ax.set_ylabel("Number of
    titles")
plt.xticks(rotation=45, ha="right"); plt.tight_layout(); plt.savefig("f4_ratings.png", dpi=170);
    plt.close()
```

```
# rating by type
ct = pd.crosstab(df["rating"], df["type"])
top_r = rc.head(8).index
fig, ax = plt.subplots(figsize=(8,4.5))
ct.loc[top_r].plot(kind="bar", color=[RED, DARK], ax=ax)
ax.set_title("Top 8 Ratings by Content Type"); ax.set_xlabel("Rating"); ax.set_ylabel("Number of titles")
plt.xticks(rotation=0); plt.tight_layout(); plt.savefig("f5_rating_type.png", dpi=170); plt.close()

# maturity group
mp = {"TV-MA":"Adults","R":"Adults","NC-17":"Adults","TV-14":"Teens","PG-13":"Teens","TV-PG":"Older
    Kids","PG":"Older Kids","TV-Y7":"Kids","TV-Y7-FV":"Kids","TV-Y":"Kids","TV-G":"Kids","G":"Kids"}
df["audience"] = df["rating"].map(mp).fillna("Other/Unrated")
S["audience"] = df["audience"].value_counts().to_dict()

# release years
ry = df["release_year"]
S["ry_min"], S["ry_max"] = int(ry.min()), int(ry.max())
S["ry_mean"] = round(float(np.mean(ry)),1); S["ry_median"] = float(np.median(ry))
yc = ry.value_counts().sort_index()
S["peak_year"] = int(yc.idxmax()); S["peak_count"] = int(yc.max())
S["share_since_2015"] = round(float((ry>=2015).mean()*100),1)
S["decade_counts"] = df["decade"].value_counts().sort_index().to_dict()
yt = df[df.release_year>=1995].groupby(["release_year","type"]).size().unstack(fill_value=0)
fig, ax = plt.subplots(figsize=(9,4.5))
ax.plot(yt.index, yt["Movie"], color=RED, lw=2.2, marker="o", ms=3, label="Movie")
ax.plot(yt.index, yt["TV Show"], color=DARK, lw=2.2, marker="o", ms=3, label="TV Show")
ax.set_title("Titles by Release Year (1995 onwards)"); ax.set_xlabel("Release year");
    ax.set_ylabel("Number of titles"); ax.legend()
plt.tight_layout(); plt.savefig("f6_release.png", dpi=170); plt.close()

# decade bar
dc = df["decade"].value_counts().sort_index()
dc = dc[dc.index>=1940]
fig, ax = plt.subplots(figsize=(8,4))
sns.barplot(x=dc.index.astype(int).astype(str)+"s", y=dc.values, color=RED, ax=ax)
ax.set_title("Titles by Release Decade"); ax.set_xlabel("Decade"); ax.set_ylabel("Number of titles")
plt.tight_layout(); plt.savefig("f7_decade.png", dpi=170); plt.close()

# heatmap genre x decade
top8 = gc.head(8).index
g2 = g[g["genres"].isin(top8) & (g.release_year>=1990)].copy()
g2["decade"] = g2["decade"].astype(int).astype(str)+"s"
hm = pd.crosstab(g2["genres"], g2["decade"]).loc[top8]
fig, ax = plt.subplots(figsize=(8,4.8))
sns.heatmap(hm, annot=True, fmt="d", cmap="Reds", linewidths=.5, ax=ax)
ax.set_title("Top 8 Genres by Release Decade (1990s onwards)"); ax.set_xlabel("Decade");
    ax.set_ylabel("")
plt.tight_layout(); plt.savefig("f8_heatmap.png", dpi=170); plt.close()

# movie duration (NumPy)
mv = df[df.type=="Movie"].copy()
mv["mins"] = mv["duration"].str.extract(r"(\d+)").astype(float)
d = mv["mins"].dropna().values
S["dur_mean"]=round(float(np.mean(d)),1); S["dur_median"]=float(np.median(d));
    S["dur_std"]=round(float(np.std(d)),1)
S["dur_min"]=int(d.min()); S["dur_max"]=int(d.max()); S["dur_p25"],S["dur_p75"]=[float(x) for x in
    np.percentile(d,[25,75])]
fig, ax = plt.subplots(figsize=(8,4))
sns.histplot(d, bins=30, color=RED, ax=ax); ax.axvline(np.mean(d), color=DARK, ls="--", label=f"Mean =
    {np.mean(d):.0f} min")
ax.set_title("Distribution of Movie Duration"); ax.set_xlabel("Duration (minutes)"); ax.legend()
plt.tight_layout(); plt.savefig("f9_duration.png", dpi=170); plt.close()
```

```
ts = df[df.type=="TV Show"]["duration"].str.extract(r"(\d+)")[0].astype(int)
S["tv_seasons"] = ts.value_counts().sort_index().head(6).to_dict()
S["tv_one_season_pct"] = round(float((ts==1).mean()*100),1)

# added per year
ya = df["year_added"].dropna().astype(int).value_counts().sort_index()
ya = ya[ya.index>=2008]
S["added_per_year"] = ya.to_dict()
fig, ax = plt.subplots(figsize=(8,4))
sns.barplot(x=ya.index, y=ya.values, color=RED, ax=ax)
ax.set_title("Titles Added to Netflix per Year"); ax.set_xlabel("Year added"); ax.set_ylabel("Number of
    titles")
plt.xticks(rotation=45); plt.tight_layout(); plt.savefig("f10_added.png", dpi=170); plt.close()

# countries
cc = df[df.country!="Unknown"]["country"].str.split(", ").explode().value_counts().head(5)
S["top_countries"] = cc.to_dict()
print(json.dumps(S, indent=1, default=str))
json.dump(S, open("stats.json","w"), default=str)
```